Lab Programs

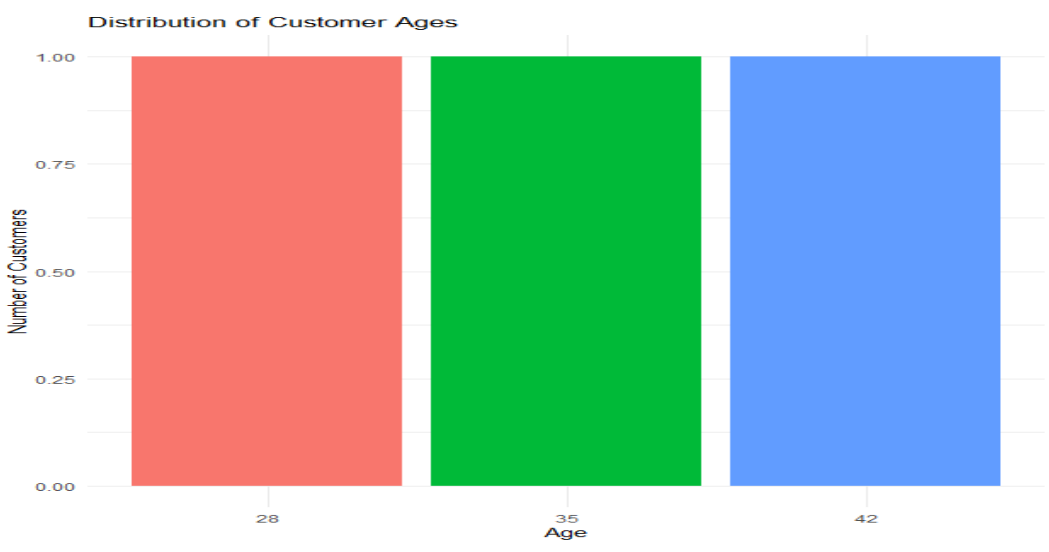
SET 16- Customer Demographics Analysis

16A. Line Chart

Program:

```
library(ggplot2)
customer_data <- data.frame(
  CustomerID = c(1,2,3),
  Age = c(28,35,42),
  Gender = c("Female","Male","Female"),
  Income = c(50000,60000,75000)
)
ggplot(customer_data, aes(x = factor(Age), fill = factor(Age))) +
  geom_bar() +
  labs(
    title = "Distribution of Customer Ages",
    x = "Age",
    y = "Number of Customers"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```

Output:

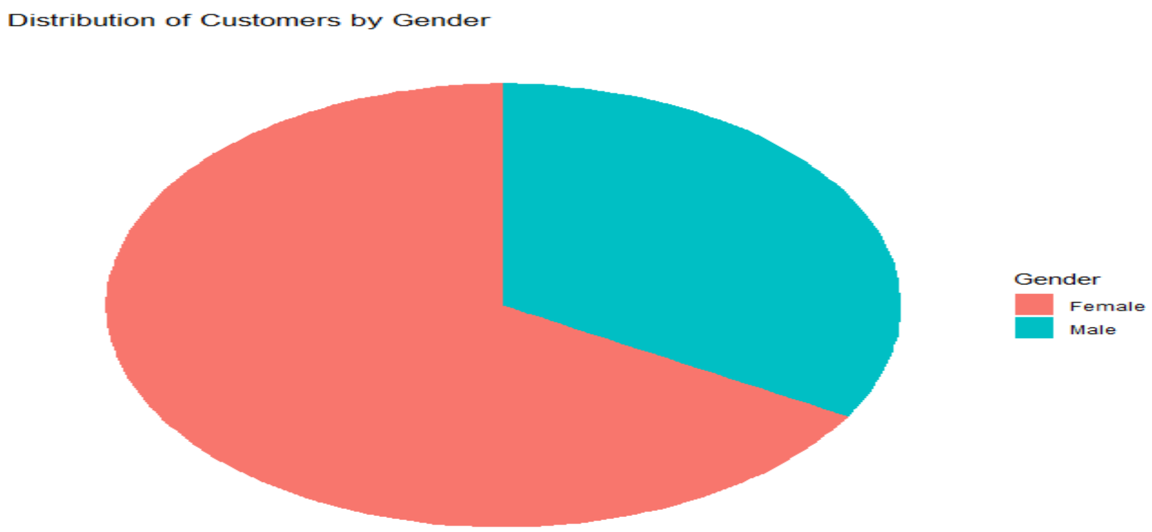


16B. Bar Chart

Program:

```
library(ggplot2)
customer_data <- data.frame(
  CustomerID = c(1,2,3),
  Age = c(28,35,42),
  Gender = c("Female","Male","Female"),
  Income = c(50000,60000,75000)
)
gender_data <- as.data.frame(table(customer_data$Gender))
colnames(gender_data) <- c("Gender","Count")
ggplot(gender_data, aes(x = "", y = Count, fill = Gender)) +
  geom_bar(stat = "identity", width = 1) +
  coord_polar(theta = "y") +
  labs(title = "Distribution of Customers by Gender") +
  theme_void()
```

Output:



16C. Table

Program:

```
View(customer_data)
```

Output:

Data: customer_data

	CustomerID	Age	Gender	Income
1	1	28	Female	50000
2	2	35	Male	60000
3	3	42	Female	75000

16D. Tableau Dashboard

Program:

```
CustomerID,Age,Gender,Income
1,28,Female,50000
2,35,Male,60000
3,42,Female,75000
```

Output:

CustomerI	Age	Gender	Income
1	28	Female	50000
2	35	Male	60000
3	42	Female	75000

SET 17- Employee Performance Analysis

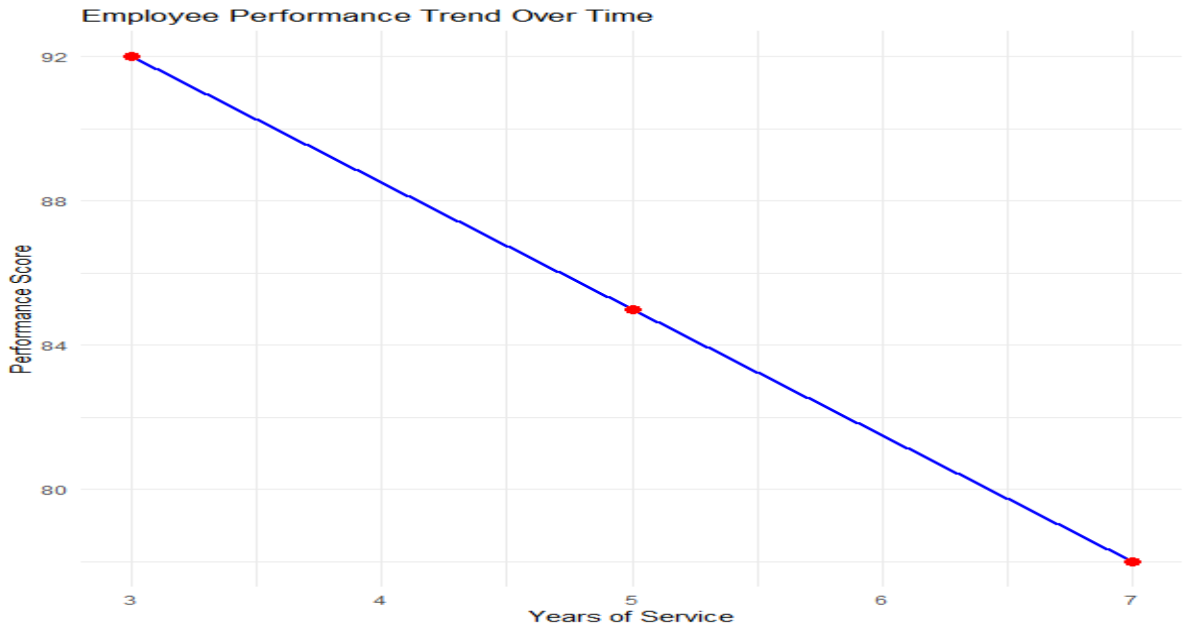
17A. Line Chart

Program:

```
library(ggplot2)
employee_data <- data.frame(
  EmployeeID = c(1,2,3),
  Department = c("Sales","HR","Marketing"),
  YearsOfService = c(5,3,7),
  PerformanceScore = c(85,92,78)
)
ggplot(employee_data,
       aes(x = YearsOfService, y = PerformanceScore, group = 1)) +
```

```
  geom_line(color = "blue", linewidth = 1) +
  geom_point(size = 3, color = "red") +
  labs(
    title = "Employee Performance Trend Over Time",
    x = "Years of Service",
    y = "Performance Score"
  ) +
  theme_minimal()
```

Output:



17B. Bar Chart

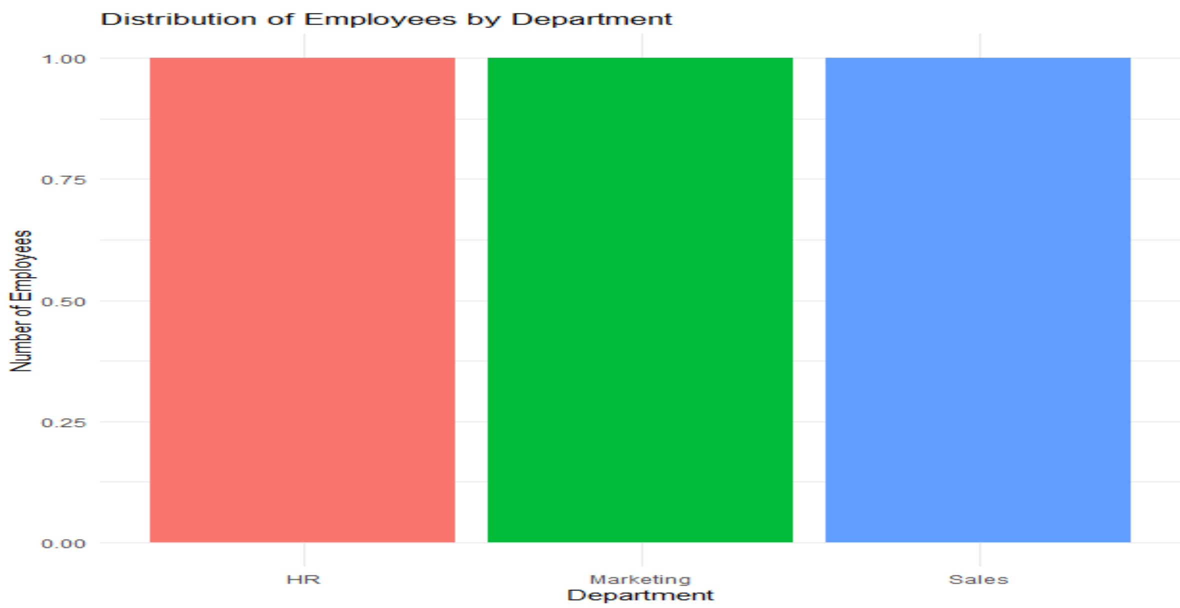
Program:

```
library(ggplot2)
ggplot(employee_data,
       aes(x = Department, fill = Department)) +
  geom_bar() +
  labs(
    title = "Distribution of Employees by Department",
    x = "Department",
    y = "Number of Employees"
  ) +
```

theme_minimal() +

theme(legend.position = "none")

Output:



17C. Table

Program:

View(employee_data)

Output:

Data: employee_data

	EmployeeID	Department	YearsOfService	PerformanceScore
1	1	Sales	5	85
2	2	HR	3	92
3	3	Marketing	7	78

17D. Tableau Dashboard

Program:

```
write.csv(employee_data,
          "Employee_Performance.csv",
          row.names = FALSE)
```

Output:

Employeel	Departmer	YearsOfSe	PerformanceScore	
1	Sales	5	85	
2	HR	3	92	
3	Marketing	7	78	

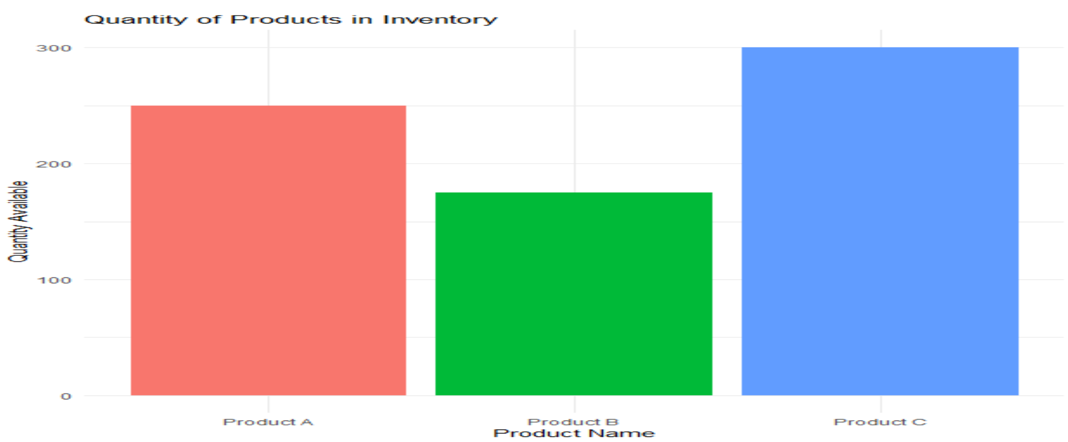
SET 18- Product Inventory Management

18A. Bar Chart

Program:

```
library(ggplot2)
inventory_data <- data.frame(
  ProductID = c(1,2,3),
  ProductName = c("Product A","Product B","Product C"),
  QuantityAvailable = c(250,175,300),
  Price = c(20,15,18)
)
ggplot(inventory_data,
       aes(x = ProductName, y = QuantityAvailable, fill = ProductName)) +
  geom_bar(stat = "identity") +
  labs(
    title = "Quantity of Products in Inventory",
    x = "Product Name",
    y = "Quantity Available"
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```

Output:

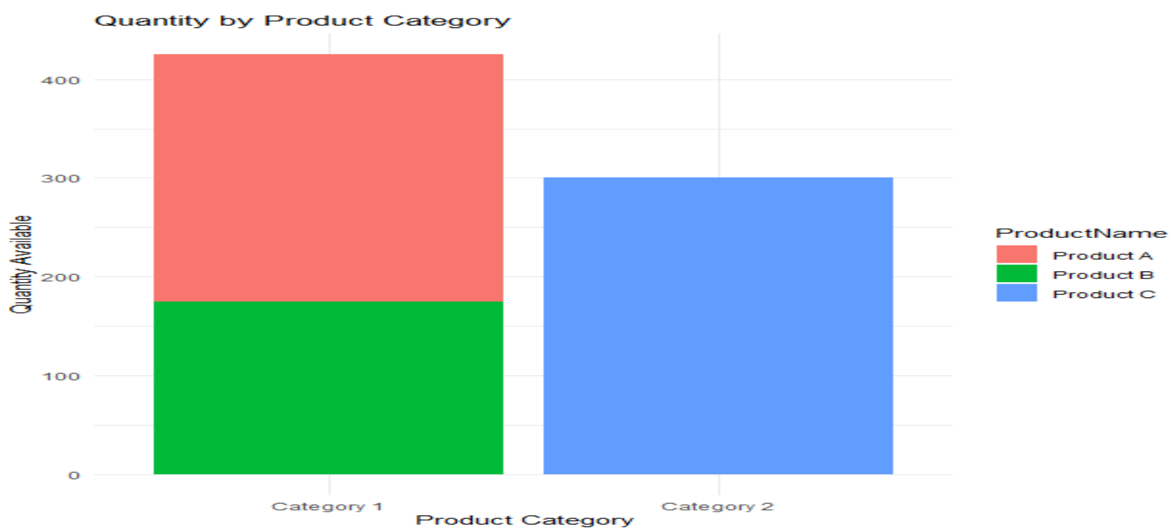


18B. Stacked Bar Chart

Program:

```
library(ggplot2)
inventory_data$Category <- c("Category 1","Category 1","Category 2")
ggplot(inventory_data,
    aes(x = Category, y = QuantityAvailable, fill = ProductName)) +
 geom_bar(stat = "identity") +
 labs(
  title = "Quantity by Product Category",
  x = "Product Category",
  y = "Quantity Available"
 ) +
 theme_minimal()
```

Output:



18C. Bar Chart

Program:

```
View(inventory_data)
```

Output:

Data: inventory_data

	ProductID	ProductName	QuantityAvailable	Price	Category
1	1	Product A	250	20	Category 1
2	2	Product B	175	15	Category 1
3	3	Product C	300	18	Category 2

18D. Tableau Dashboard

Program:

```
write.csv(
  inventory_data,
  "Product_Inventory.csv",
  row.names = FALSE
)
```

Output:

ProductID	ProductNa	QuantityAv	Price	Category
1	Product A	250	20	Category 1
2	Product B	175	15	Category 1
3	Product C	300	18	Category 2

SET 19- Survey Responses Analysis

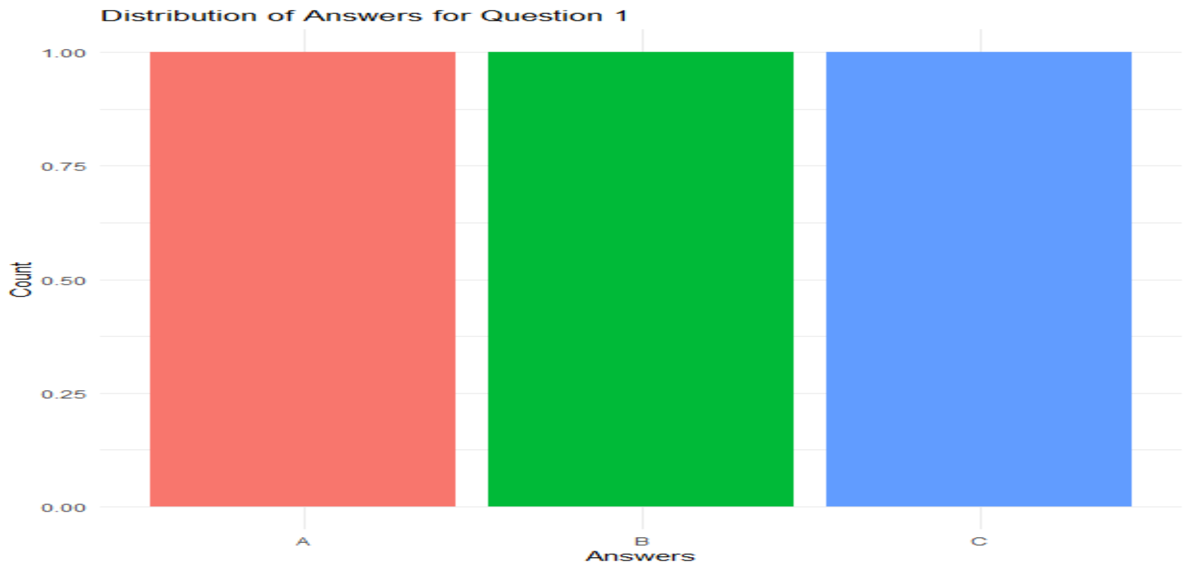
19A. Grouped Bar Chart

Program:

```
library(ggplot2)
survey_data <- data.frame(
  SurveyID = c(1,2,3),
  Question1 = c("A","B","C"),
  Question2 = c("B","A","A"),
  Question3 = c("C","D","B")
)
q1_data <- as.data.frame(table(survey_data$Question1))
colnames(q1_data) <- c("Answer","Count")
ggplot(q1_data, aes(x = Answer, y = Count, fill = Answer)) +
  geom_bar(stat = "identity") +
  labs(
    title = "Distribution of Answers for Question 1",
    x = "Answers",
    y = "Count"
```

```
  ) +
  theme_minimal() +
  theme(legend.position = "none")
```

Output:



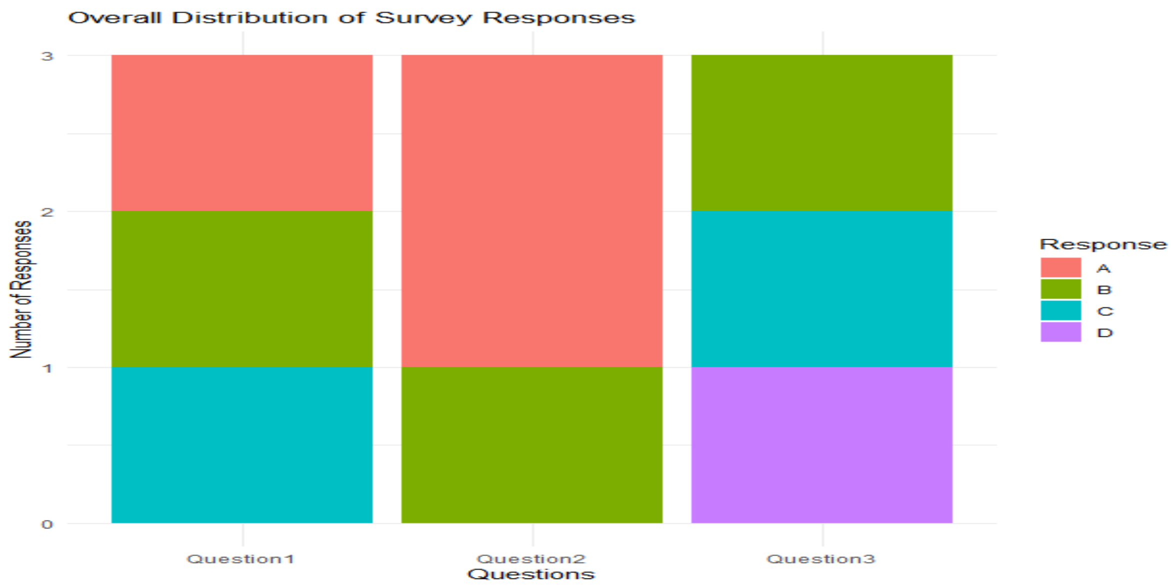
19B. Stacked Bar Chart

Program:

```
library(ggplot2)
library(tidyr)
survey_long <- pivot_longer(
  survey_data,
  cols = starts_with("Question"),
  names_to = "Question",
  values_to = "Response"
)
ggplot(survey_long,
       aes(x = Question, fill = Response)) +
  geom_bar() +
  labs(
    title = "Overall Distribution of Survey Responses",
    x = "Questions",
```

```
    y = "Number of Responses"
  ) +
  theme_minimal()library(ggplot2)
library(tidyr)
survey_long <- pivot_longer(
  survey_data,
  cols = starts_with("Question"),
  names_to = "Question",
  values_to = "Response"
)
ggplot(survey_long,
       aes(x = Question, fill = Response)) +
  geom_bar() +
  labs(
    title = "Overall Distribution of Survey Responses",
    x = "Questions",
    y = "Number of Responses"
  ) +
  theme_minimal()
```

Output:



19C. Table

Program:

```
View(survey_data)
```

Output:

Data: survey_data

	SurveyID	Question1	Question2	Question3
1	1	A	B	C
2	2	B	A	D
3	3	C	A	B

19D. Tableau Dashboard

Program:

```
write.csv(
  survey_data,
  "Survey_Responses.csv",
  row.names = FALSE
)
```

Output:

SurveyID	Question1	Question2	Question3
1	A	B	C
2	B	A	D
3	C	A	B

SET 20- Stock Analysis

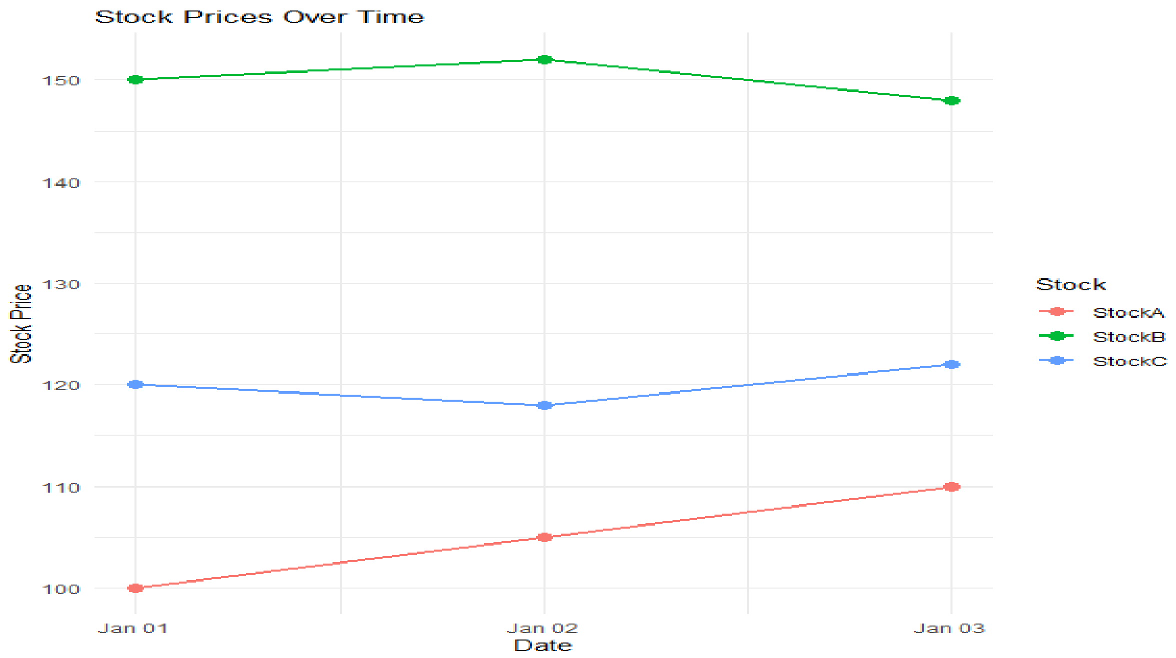
20A. Line Chart

Program:

```
library(ggplot2)
library(tidyr)
stock_data <- data.frame(
  Date = as.Date(c("2023-01-01","2023-01-02","2023-01-03")),
  StockA = c(100,105,110),
  StockB = c(150,152,148),
```

```
  StockC = c(120,118,122)
)
stock_long <- pivot_longer(
  stock_data,
  cols = c(StockA, StockB, StockC),
  names_to = "Stock",
  values_to = "Price"
)
ggplot(stock_long,
       aes(x = Date, y = Price, color = Stock, group = Stock)) +
  geom_line(linewidth = 1) +
  geom_point(size = 3) +
  labs(
    title = "Stock Prices Over Time",
    x = "Date",
    y = "Stock Price"
  ) +
  theme_minimal()
```

Output:

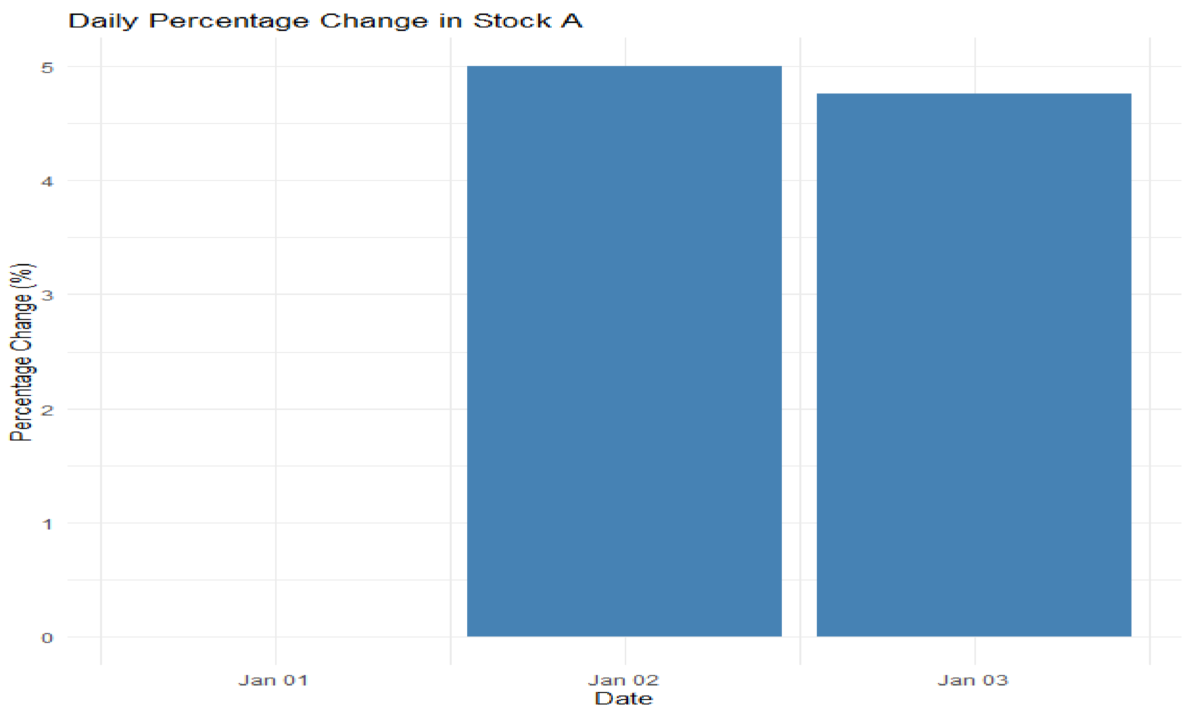


20B. Bar Chart

Program:

```
library(ggplot2)
stock_data$PercentageChange <- c(
  0,
  (105 - 100) / 100 * 100,
  (110 - 105) / 105 * 100
)
ggplot(stock_data,
       aes(x = Date, y = PercentageChange)) +
  geom_bar(stat = "identity", fill = "steelblue") +
  labs(
    title = "Daily Percentage Change in Stock A",
    x = "Date",
    y = "Percentage Change (%)"
  ) +
  theme_minimal()
```

Output:



20C. Table

Program:

```
View(stock_data)
```

Output:

Data: stock_data

	Date	StockA	StockB	StockC	PercentageChange
1	2023-01-01	100	150	120	0.000000
2	2023-01-02	105	152	118	5.000000
3	2023-01-03	110	148	122	4.761905

20D. Tableau Dashboard

Program:

```
write.csv(
  stock_data,
  "Stock_Prices.csv",
  row.names = FALSE
)
```

Output:

Date	StockA	StockB	StockC	PercentageChange
########	100	150	120	0
########	105	152	118	5
########	110	148	122	4.761905